Spatiotemporal Coverage of a Moon-Based Synthetic Aperture Radar: Theoretical Analyses and Numerical Simulations

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***Abstract*— The spatiotemporal coverage of a Moon-based synthetic aperture radar (SAR) is analyzed based on the imaging geometry, upon which the spatial coverage and image formulation rely. The distance from the Earth to the Moon-based SAR and bounds of the grazing and azimuthal angles jointly determine the coverage area on the Earth's surface. Meanwhile, the ground coverage of the Moon-based SAR is determined by the bounds of the grazing and azimuthal angles and geographic coordinates of the nadir point at a specified time. Moreover, the temporal variation in the spatial coverage is pertinent to the temporally varying nadir point of the Moon-based SAR on the Earth's surface. Furthermore, numerical simulations using the lunar ephemeris data are carried out to complement the analysis and to illustrate the spatiotemporal coverage. Finally, a guideline for the optimal site selection of a Moon-based SAR is proposed. In conclusion, a Moon-based SAR has the potential to perform long-term, continuous Earth observations on a global scale to enhance our capability to understand the planet.**

***Index Terms*— Azimuthal angle, coverage area, grazing angle, ground coverage, imaging geometry, Moon-based synthetic aperture radar (SAR), optimal site selection, spatiotemporal coverage.**

I. Introduction

With the renewed international interest in the exploration of the Moon in recent years [1]–[3], a series of exploration missions to the Moon are implemented and proposed by the scientific community [4]–[8]. Such tasks are of great scientific significance and tremendous application values [9], [10], including using the Moon as a remote sensing platform for the Earth observation [11]–[16]. A Moon-based synthetic aperture radar (SAR) can, by different means, monitor global environmental phenomena. Therefore, the Moon-based SAR would be an attractive complement to the current low Earth orbit SAR (LEOSAR) systems [17]–[25].

For Earth observation, the coverage performance is of great significance. Detailed analyses of the Moon-based sensor's coverage, however, mainly concentrate on the optical sensors [26]–[28]; few of these studies concern the coverage of a Moon-based SAR. It is commonly known that high orbit SAR systems can potentially offer extensive spatial coverage within a relatively short revisit period [29]. Hence, it is vital to have an SAR system bearing higher orbital altitude to obtain continuous observations in both space and time. For example, Earth observations from the LEO SAR systems are quite limited, with revisit times measured in days to weeks and swath widths ranging from dozens to hundreds of kilometers [19]. On the contrary, the Moon-based SAR can achieve a wider swath width, whose size can be up to more than thousands of kilometers, with a relatively shorter revisit period [25]. In addition, the Moon-based SAR can provide consecutive global coverage during the observation [19].

Some preliminary analyses on the Moon-based SAR's coverage are carried out in [19] and [25]. However, Guo *et al.* [19] simply put forward an idea of the potential of the Moon-based SAR for observing the Earth in accordance with a simple model. Besides, there is a lack of theoretical analysis about the coverage in line with spatial and temporal variations [25]. Furthermore, it is noted that the analysis of the LEO SAR coverage is not valid for the Moon-based SAR, for the extensive spatial coverage of the Moon-based SAR is greatly affected by the Earth's curvature. Accordingly, it is required to examine the spatiotemporal coverage of the Moon-based SAR.

In this article, we aim to explore the spatiotemporal coverage of the Moon-based SAR, to study the potential of the Moon-based SAR for monitoring the large-scale geoscience phenomena, and to propose guidelines for the optimal site selection of the Moon-based SAR. From the imaging geometry of the Moon-based SAR, the spatial coverage in terms of the coverage area and ground coverage range at a specified time is analyzed. In particular, the temporal variation in the spatial coverage pertaining to the temporally varying nadir point of the Moon-based SAR is examined. It follows that the spatiotemporal coverage of the Moon-based SAR is simulated

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based on the lunar ephemeris data from the development ephemeris 430 (DE430) produced by Jet Propulsion Laboratory [30]. Finally, following the simulation results, a guideline for the optimal site selection of the Moon-based SAR is provided.

The remainder of this article is organized as follows. In Section II, the coordinate systems and their transformations used for depicting the position of the Moon-based SAR are introduced. This yields key parameters related to the coverage performance of the Moon-based SAR. Afterward, the spatial coverage in terms of the coverage area and ground coverage range at a specified time and the temporal variation in the spatial coverage concerning the temporally varying nadir point are analyzed in Section III. In Section IV, numerical simulations based on the lunar ephemeris are performed to complement theoretical analyses and to illustrate the spatiotemporal coverage of the Moon-based SAR. Then, a guideline is proposed to select the optimal site of the Moon-based SAR. Finally, a conclusion is drawn.

II. Imaging Geometric Model of the Moon-Based SAR

To understand the Moon-based SAR's imaging geometry, it is critical to obtain the accurate position vectors of the ground target and SAR system under a spatial reference system in a specified time coordinate [22], [27]. The barycentric dynamical time (TDB) is used for providing the coordinate time scale in this article [23], [30]. Then, the Moon-based SAR and the ground target are mapped to the same reference frame through a series of coordinate conversions, to be elaborated next.

A. Definitions and Transformations of Coordinate Reference Frames

To uniquely describe the locations of the SAR system on the Moon's surface and the target on the Earth's surface, we introduce four reference frames (see Fig. 1):

1) Moon-Centered Moon-Fixed Coordinate System (MCMF): The MCMF is used for providing the location of the SAR system on the lunar surface. It is a right-handed reference frame where the x-axis extends from the Moon's center to the mean Earth direction. The z-axis points to the mean rotational pole of the Moon.

2) Moon-Centered Inertial Coordinate System (MCI): The origin of the MCI is defined as the Moon's center with the z-axis along the direction of the celestial north and the x-axis toward the vernal equinox. The y-axis obeys a right-hand rule.

3) Earth-Centered Inertial Coordinate System (ECI): It is a right-handed reference frame and is served to describe the location of the Moon along its orbit. In the ECI, its origin is located at the Earth's center, with the x-axis pointing to the vernal equinox and the z-axis pointing to the celestial north.

4) Earth-Centered Rotational Coordinate System (ECR): The ECR is a right-handed reference frame; it is also known as the conventional terrestrial system. The origin of the ECR is the Earth's center. The x-axis is directed toward the Greenwich Meridian. The z-axis points to the North pole of the Earth.

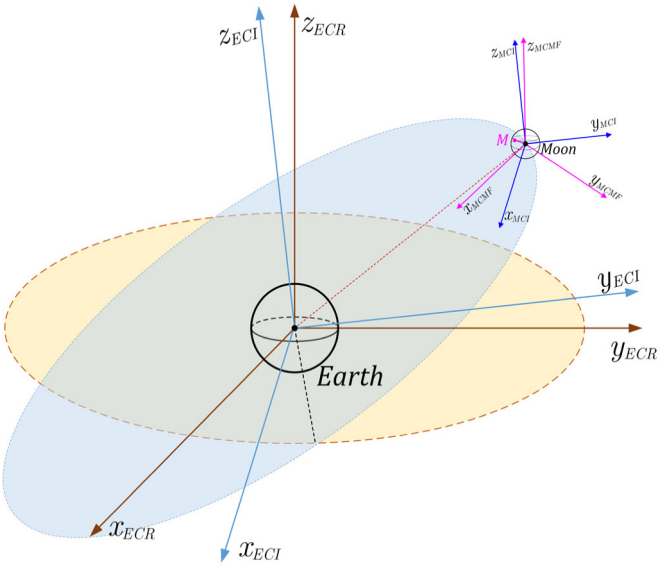


Fig. 1. Definitions of the reference frames, where "M" represents the site of the Moon-based SAR on the lunar surface. For the sake of clarity, this diagram is not to scale.

The position vector of the SAR system on the Moon's surface in the ECI can be determined by [17]

$$\mathbf{R}_{\mathrm{SAR}}^{\mathrm{ECI}} = \mathbf{M}_{\mathrm{MCI}}^{\mathrm{MCMF}} \mathbf{R}_{\mathrm{SAR}}^{\mathrm{MCMF}} + \mathbf{D}_{\mathrm{EM}}^{\mathrm{ECI}} \tag{1}$$

where $\mathbf{R}_{\mathrm{SAR}}^{\mathrm{MCMF}}$ and $\mathbf{D}_{\mathrm{EM}}^{\mathrm{ECI}}$ are the position vector of the SAR system in the MCMF and that of the Moon's center in the ECI, respectively. Generally, $\mathbf{D}_{\mathrm{EM}}^{\mathrm{ECI}}$ can be obtained from the lunar ephemeris. $\mathbf{M}_{\mathrm{MCI}}^{\mathrm{MCMF}}$, the coordinate transformation matrix from the MCMF to the MCI, is defined as [21]

$$\mathbf{M}_{MCI}^{\mathrm{MCMF}} = \mathbf{R}_z(-\phi_m)\mathbf{R}_x(-\theta_m)\mathbf{R}_z(-\psi_m) \tag{2}$$

where ϕ_m, θ_m, and ψ_m are the three auxiliary angles assigned to describe the libration of the Moon [27]. $\mathbf{R}_x$, $\mathbf{R}_y$, and $\mathbf{R}_z$ are the rotation matrices about the x-, y-, and z-axes, respectively [31]; their expressions are given in Appendix A.

Then, the position vector of the Moon-based SAR in the ECR is acquired through the following transformation [32]:

$$\mathbf{R}_{\mathrm{SAR}}^{\mathrm{ECR}} = \mathbf{U}_{\mathrm{ECR}}^{\mathrm{ECI}} \mathbf{R}_{\mathrm{SAR}}^{\mathrm{ECI}} \tag{3}$$

with the transformation matrix expressed as

$$\mathbf{U}_{\mathrm{ECR}}^{\mathrm{ECI}} = \Pi\Theta\mathbf{N}\mathbf{P} \tag{4}$$

where matrices $\mathbf{P}$, $\mathbf{N}$, Θ, and Π represent the coordinate changes due to the precession, nutation, Earth rotation, and polar motion, respectively. The complete transformations from the ECI to the ECR can be found in Appendix A.

Once the coordinate transformation is accomplished, the distance from the Earth to the Moon-based SAR, the nadir point's latitude, and the nadir point's longitude can be obtained, respectively, by

$$R_{\mathrm{SAR}} = \sqrt{x_m^2 + y_m^2 + z_m^2} \tag{5}$$

$$\delta_m = \tan^{-1}\left[z_m/(x_m^2 + y_m^2)^{0.5}\right] \tag{6}$$

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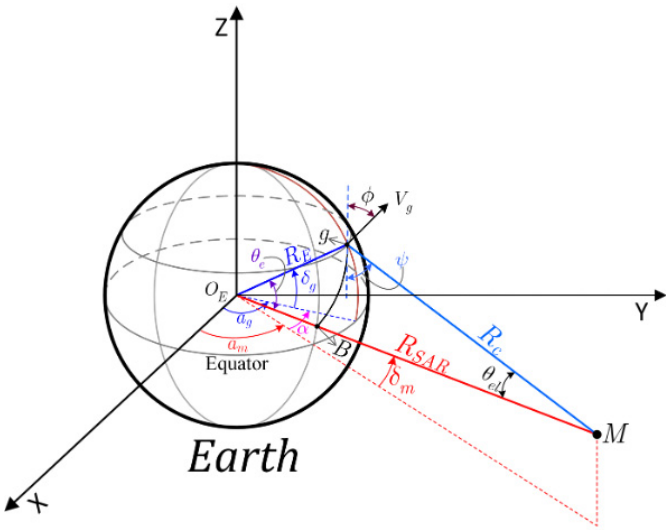


Fig. 2. Sketch of the location of the Moon-based SAR in the ECR. For the sake of clarity, this diagram is not to scale.

$$a_m = \begin{cases} \tan^{-1}(y_m/x_m), & x \geq 0 \\ \tan^{-1}(y_m/x_m) + \pi, & x < 0, y \leq 0 \\ \tan^{-1}(y_m/x_m) - \pi, & x < 0, y > 0 \end{cases} \quad (7)$$

where (x_m, y_m, z_m) are the coordinates of the Moon-based SAR in the ECR. As the Moon-based SAR is still at the phase of conceptual design, the specific site of the Moon-based SAR is yet to be determined [25], [33]. To proceed with this analysis, the position of the Moon-based SAR is assumed to be the same as that of the Moon because of the extremely long distance from the Earth to the Moon.

B. Parameters Related to the Coverage Performance

Fig. 2 describes the location of the Moon-based SAR in the ECR. The explanations of symbols that appear in this reference frame are listed in Table I. We derive the parameters related to the coverage performance of the Moon-based SAR as given below.

The geocentric angle, defined as the angle between the nadir point of the Moon-based SAR and the ground target referring to the Earth's center, is

$$\theta_e = \cos^{-1}(\cos\delta_m \cos\delta_g \cos\alpha + \sin\delta_m \sin\delta_g) \quad (8)$$

where α is the longitudinal deviation and is defined as

$$\alpha = a_g - a_m. \quad (9)$$

The elevation angle that is the angle between the Moon-based SAR's position vector and the slant range vector is specified by

$$\theta_{el} = \cos^{-1}\{[R_{\mathrm{SAR}} - R_E(\cos\delta_m \cos\delta_g \cos\alpha + \sin\delta_m \sin\delta_g)]/R_c\}. \quad (10)$$

TABLE I
PARAMETERS OF THE MOON-BASED SAR IN THE ECR

Symbol	Parameters	Symbol	Parameters
M	Site of Moon-based SAR	B	Nadir point
g	Ground target	R_E	Earth's radius at g
a_g	Earth's longitude at g	a_m	Nadir point's longitude
δ_g	Earth's latitude at g	δ_m	Nadir point's latitude
R_{SAR}	Earth–Moon-based SAR distance	R_c	Slant range of the Moon-based SAR
V_g	Linear velocity of the Earth's rotation at g	ψ	Grazing angle
θ_e	Geocentric angle	θ_{el}	Elevation angle
ϕ	Azimuthal angle	α	Longitudinal deviation

The grazing angle, the angle between the slant range and the ground range plane, can be expressed as

$$\psi = \sin^{-1}\{[R_{\mathrm{SAR}}(\cos\delta_m \cos\delta_g \cos\alpha + \sin\delta_m \sin\delta_g) - R_E]/R_c\}. \quad (11)$$

The azimuthal angle, the angle between the slant range vector in the ground plane and the velocity vector of the Earth's self-rotation (eastward), takes the form of

$$\phi = \cos^{-1}\{\cos\delta_m \sin\alpha[1 - (\cos\delta_g \cos\delta_m \cos\alpha + \sin\delta_g \sin\delta_m)^2]^{-0.5}\}. \quad (12)$$

It should be noted that the direction of the Earth's self-rotation is used as a reference, represented by an azimuthal angle of 0°.

R_c, the slant range of the Moon-based SAR, is given by (13), as shown at the bottom of this page.

Detailed derivations of (9)–(13) are given in Appendix B.

The geocentric and azimuthal angles, together with the grazing angle or elevation angle, determine the spatial coverage of a Moon-based SAR. Here, we choose the geocentric, azimuthal, and grazing angles to examine the Moon-based SAR's spatial coverage.

According to the Moon-based SAR's imaging geometry in Fig. 3, the grazing angle should be bounded by

$$\psi_{\mathrm{far}} \leq \psi \leq \psi_{\mathrm{near}} \quad (14)$$

where ψ_{far} and ψ_{near} are the lower and the upper bounds of the grazing angle, respectively.

Substituting (11) into (8) and applying (14), we can rewrite (8) to

$$\cos\theta_e(\delta_m, \delta_g, \alpha) = \sin\psi \cdot R_c/R_{\mathrm{SAR}} + R_E/R_{\mathrm{SAR}}. \quad (15)$$

As far as the Moon-based SAR's spatial coverage is concerned, the slant range in (13) can be further approximated to

$$R_{ca} \approx R_{\mathrm{SAR}}\left\{1 + [0.5R_E^2 - R_{\mathrm{SAR}}R_E \cos\theta_e(\delta_m, \delta_g, \alpha)]/R_{\mathrm{SAR}}^2\right\}. \quad (16)$$

$$R_c = \sqrt{R_{\mathrm{SAR}}^2 + R_E^2 - 2R_{\mathrm{SAR}}R_E(\cos\delta_m \cos\delta_g \cos\alpha + \sin\delta_m \sin\delta_g)} \quad (13)$$

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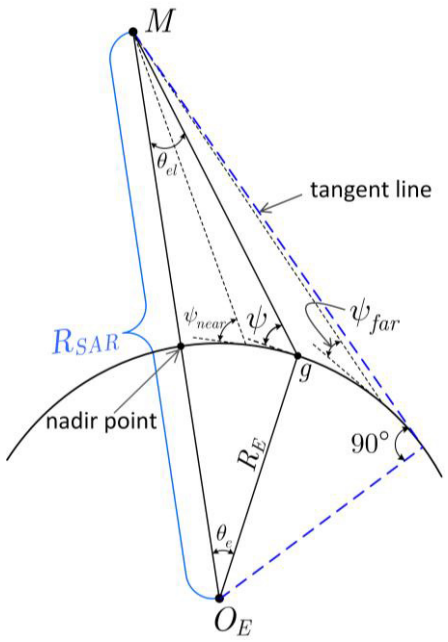


Fig. 3. Imaging geometry of the Moon-based SAR, ψ_{far} and ψ_{near} are the lower and the upper bounds of the grazing angle, respectively. This diagram is not to scale.

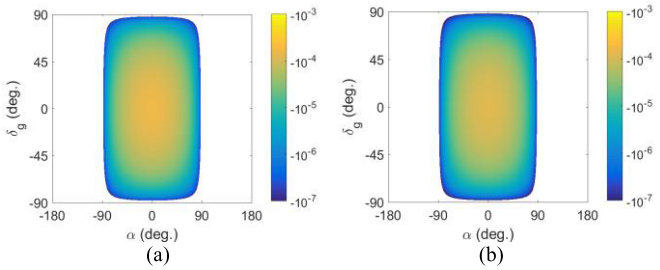


Fig. 4. Relative errors δR in the $\alpha - \delta_g$ domain at (a) perigee of the lunar orbit and (b) apogee of the lunar orbit.

To confirm the validity of (16) for analyzing the Moon-based SAR's spatial coverage, a relative error is defined as

$$\delta R = (R_c - R_{ca})/R_{\text{SAR}}. \tag{17}$$

Fig. 4 presents the relative errors at the perigee and the apogee of the lunar orbit in the $\alpha - \delta_g$ domain. For the sake of simplicity but without loss of generality, the latitude of the Moon-based SAR's nadir point is set to 0°, and the grazing angle is ranged from 0° to 90°. As detailed in Fig. 4, the maximum value of the relative error δR is far smaller than 10^{-3}, which appears at the nadir point regardless of the distance from the Earth to the Moon-based SAR. This suggests that the approximation of the slant range in (16) has little bearing on the dimensionless quantity, δR. In other words, the error induced by the approximation of the slant range can be reasonably ignored in determining the spatial coverage of a Moon-based SAR. However, we must compensate for this error given in the image focusing. Accordingly, (15) can be approximated by taking account of (16) to

$$\cos\theta_e(\delta_m, \delta_g, \alpha) \approx \kappa(\psi, R_{\text{SAR}}) \tag{18}$$

with

$$\kappa(\psi, R_{\text{SAR}}) = \frac{\sin\psi + 0.5R_E/R_{\text{SAR}}}{\sin\psi \times R_E/R_{\text{SAR}} + 1} + 0.5R_E/R_{\text{SAR}}. \tag{18a}$$

Note that the ratio of the Earth's radius to the Earth–Moon-based SAR distance is spatiotemporally varying as follows:

$$R_E/R_{\text{SAR}} \approx R_E^A/R_{\text{SAR}}^A \times \left(1 - R_{\text{SAR}}^T/R_{\text{SAR}}^A + R_E^S/R_E^A\right). \tag{19}$$

where R_E^S is the spatially varying component of the Earth's radius at an arbitrary position on the Earth's surface with respect to the average Earth's radius, R_E^A; R_{SAR}^T is the temporally varying part of the Earth–Moon-based SAR distance at any time referring to the average distance from the Earth to the Moon-based SAR, R_{SAR}^A.

According to [34] and [35], at the perigee, the Moon is 356 410 km from the Earth; at the apogee, it is 406 697 km from the Earth, and the average Earth–Moon distance is 385 000 km. In contrast, the Earth's radius ranges from 6356.7 to 6378.1 km, with an average value of 6371.0 km [36]. It can be calculated that $R_{\text{SAR}}^T/R_{\text{SAR}}^A$ is approximately 30 times larger than R_E^S/R_E^A during one period of the lunar revolution. Thus, the spatially varying component of the Earth's radius contributes little to (19) (i.e., the ratio of the Earth's radius to the Earth–Moon-based SAR distance). Hence, we can reasonably accept a constant Earth's radius (6371.0 km) in this study.

III. Theoretical Analyses of the Spatiotemporal Coverage of a Moon-Based SAR

For a passive remote sensing sensor placed on the lunar surface, its spatial coverage can be obtained by letting $\psi_{\text{far}} = 0°$ and $\psi_{\text{near}} = 90°$. As to the Moon-based SAR system, the spatial coverage should take the system performance and image quality into account. Thus, the effective range of the Moon-based SAR's spatial coverage is first studied.

A. Effective Range of the Moon-Based SAR's Spatial Coverage

The Moon can be regarded as a longstanding and stable platform, on which a set of antennas or array can be installed for Earth observation. Generally, a single antenna can observe a limited portion of the Earth's surface in a given time frame. However, multiantennas or antenna array can achieve the maximum extent of the spatial coverage by adjusting the pointing direction of each antenna. For example, the half-power beamwidth of a single antenna is [37]

$$\beta \approx k_{xy}\lambda/\ell \tag{20}$$

where λ is the wavelength, ℓ is the aperture length, and k_{xy} is the aperture-illumination taper factor for the antenna. Generally, a value of $k_{xy} = 1.5$ is appropriate for an antenna with high beam efficiency. Some system parameters of the Moon-based SAR had been proposed in [19], including an aperture length in the range of 57 m and a wavelength of 0.24 m. Under these configurations, the half-power beamwidth of the tapered wave can be up to 0.362°. On the other hand, the entire Earth disk in the range direction is less than 2°

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when observing from the Moon's surface. Thus, six antennas with such configurations can cover the whole Earth disk in the range direction by properly adjusting the pointing direction of each antenna (ignoring the overlapped coverage region). A similar technique can be applied when the azimuth direction is observed. Accordingly, the maximum extent of the spatial coverage can be obtained using utilizing multiantennas.

However, the traditional SAR system suffers the issue of left–right ambiguity because the targets situated symmetrically about the moving trace of the radar have the same range histories. In the case of the Moon-based SAR, such an issue may not be a problem. Two sets of antenna systems consisting of multiple antennas with the same configurations can be applied for right-looking and left-looking separately simultaneously to isolate ambiguous echoes. In addition, the spatial coverage of the Moon-based SAR is subject to the imaging geometry because the SAR system cannot have nadir imaging and it may perform poorly at large incident angles [38]. Thus, the grazing angle should be limited to a valid range: The observed areas of the satellite SAR are commonly restricted within the range from 20° to 60° of the incident angles, namely from 70° to 30° of the grazing angles. Regarding the Moon-based SAR, the near- and far-grazing angles should be limited to a practical range for covering the globe within each cycle of the lunar revolution.

The far-grazing angle is determined such that both the North and South Poles of the Earth are observable during one lunar revolution. By letting $|\delta_g| = 90°$, and from (10), (15), and (18), the following relationship is readily obtained:

$$\psi_{\text{far}} \leq \sin^{-1}\left\{\frac{\sin|\delta_m| - R_E/R_{\text{SAR}}}{\left(1 + R_E^2/R_{\text{SAR}}^2 - 2\sin|\delta_m| \cdot R_E/R_{\text{SAR}}\right)^{0.5}}\right\}. \tag{21a}$$

Once the far grazing angle exceeds its upper bound, neither the North pole nor the South pole is covered by the Moon-based SAR wherever the nadir point is located. It is noted that even when the far-grazing angle is bounded, the North pole (or the South pole) is not always observed by the Moon-based SAR. The observation of the North pole (or the South pole) is still subjected to the nadir point's latitude, as will be evidenced in Section III-B.

Similarly, the near-grazing angle should satisfy the following relationship so as to cover the equatorial regions:

$$\psi_{\text{near}} \geq \sin^{-1}\left\{\frac{R_E/R_{\text{SAR}} - \cos\delta_m}{\cos\delta_m \cdot R_E/R_{\text{SAR}} - 0.5 \cdot R_E^2/R_{\text{SAR}}^2 - 1}\right\}. \tag{21b}$$

Note that only when the near-grazing angle is larger than the lower bound can the equatorial regions of the Earth be observed during each cycle of the lunar revolution.

According to [39], the inclination of the lunar orbit to the Earth's equatorial plane stands at between 18.3° and 28.6°; in other words, the maximum scale of the nadir point's latitude varies from 18.3° to 28.6°. Therefore, in Fig. 5(a), we plot the bounds of the far-grazing angle at the apogee and the perigee of the lunar orbit versus the nadir point's latitude. Then, the bounds of the near-grazing angle at the apogee and the perigee of the lunar orbit with respect to the magnitude of the nadir point's latitude are plotted in Fig. 5(b). The range of latitude for the nadir point of the Moon-based SAR is 18.3°–28.6°.

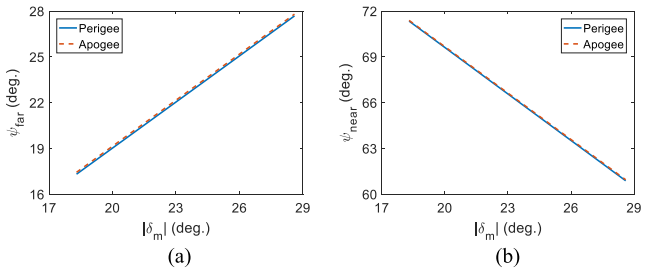


Fig. 5. With the Moon located at the apogee and the perigee of the lunar orbit (a) shows the upper bound of the far-grazing angle for covering the Earth's poles (neither the North pole nor the South pole is observed when the far-grazing angle is beyond the lines) and (b) shows the lower bound of the near-grazing angle for covering the equatorial regions (the equatorial regions will not be observed when the near-grazing angle is below the lines).

Fig. 5(a) shows that the upper bound of the far-grazing angle can reach 27.8026° at the apogee of the lunar orbit when the orbital inclination is the maximum. Nevertheless, the far-grazing angle should be limited to 17.3174° for a continuous global observation during one lunar revolution with the minimum orbital inclination. Otherwise, the revisit period of the North pole (or the South pole) can take up to tens of months or even several years, which is clearly not desirable. Similarly, as presented in Fig. 5(b), the near-grazing angle should be larger than 71.3934° to cover the equatorial regions during one period of the lunar revolution when the orbital inclination is minimum. Consequently, we can set the effective range of the grazing angle to [15°, 75°].

Even though the spatial coverage is bounded by the near- and far-grazing angles, the image quality of the Moon-based SAR is still an issue. The Moon-based SAR system usually works in the squint mode when observing the bounded regions, which would lead to pixel skewing on the ground and raise a significant challenge to the image quality [23]. Under the skewing effect, satisfactory imaging is only available when the iso-range and iso-Doppler profiles are relatively far from being parallel [31]. Generally, the quasi-parallel regions of the iso-range and iso-Doppler contours occur at where the azimuthal angles are close to 0° or 180°. Therefore, it is practical to examine the spatial coverage of the Moon-based SAR by limiting the azimuthal angle, namely

$$|\cos\phi| \leq \cos\phi_b \tag{22}$$

where ϕ_b is the bound of the azimuthal angle; here, we set it to 30° to ensure the good image quality.

In Fig. 6(a)–(d), we plot the iso-range and iso-Doppler profiles in the $\alpha - \delta_g$ domain within the boundary set by (22) with the Moon-based SAR locating at different positions listed in Table II. The beam center crossing time is set to 00:00:00 without loss of generality.

As shown in Fig. 6, the iso-Doppler and iso-range contours cut across each other wherever the Moon-based SAR is located. This suggests that the image quality is acceptable even when the Moon-based SAR system operates in a squint mode.

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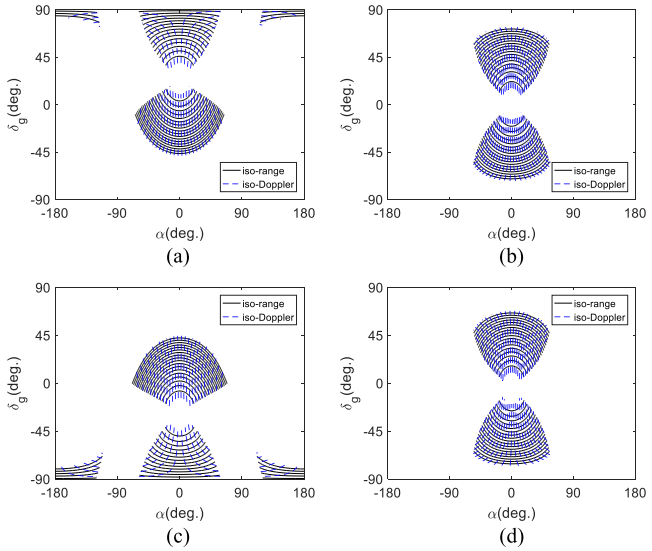


Fig. 6. Iso-Doppler and iso-range profiles in the $\alpha - \delta_g$ domain at 00:00:00 on (a) March 20, 2024, (b) March 25, 2024, (c) April 01, 2024, and (d) April 07, 2024.

TABLE II
COORDINATES OF THE MOON-BASED SAR IN THE ECR AT DIFFERENT TIMES

Time \ Coordinates	X (km)	Y(km)	Z(km)
Mar.20, 2024, 00:00:00	221041.4734	-289559.7324	167171.4715
Mar.25, 2024, 00:00:00	405558.5205	-8874.6121	4569.0584
Apr.01, 2024, 00:00:00	74979.4046	330070.1283	-183398.9317
Apr.07, 2024, 00:00:00	-332302.4149	133331.8257	-30492.2954

The latitudes of the nadir point on Mar.20 and Apr.01 are close to the maximum and minimum values, while latitudes of the nadir point on Mar.25 and Apr.07 are approaching 0° of the latitude.

Thus, the skewing effect exerts little impact on the spatial coverage of the Moon-based SAR within the scope from (22).

Finally, we can obtain the bounds of the geocentric angle that determines the Moon-based SAR's spatial coverage as

$$\theta_e^{\text{low}} \le \theta_e(\delta_m, \delta_g, \alpha) \le \theta_e^{\text{up}} \tag{23}$$

where θ_e^{up} is the upper bound of the geocentric angle, defined as

$$\theta_e^{\text{up}} = \cos^{-1}[\kappa(\psi_{\text{far}}, R_{\text{SAR}})]. \tag{24}$$

and θ_e^{low}, the lower bound of geocentric angle, is

$$\theta_e^{\text{low}} = \begin{cases} \cos^{-1}[\kappa(\psi_{\text{near}}, R_{\text{SAR}})], & |\alpha| \le \alpha_{\text{th}} \\ \cos^{-1}[\cos\phi_b(\cos^2\phi_b - \cos^2\delta_m \sin^2\alpha)^{0.5}], & |\alpha| \ge \alpha_{\text{th}} \end{cases} \tag{25}$$

where the threshold value α_{th} is given by

$$\alpha_{\text{th}} = \sin^{-1}\{\cos\phi_b \cdot \cos^{-1}\delta_m \cdot [1 - \kappa^2(\psi_{\text{near}}, R_{\text{SAR}})]^{0.5}\}. \tag{26}$$

Note that the upper bound of the geocentric angle, related to the far-grazing angle, determines the exterior boundary of the spatial coverage. In comparison, the lower bound of the geocentric angle is correlated with the near-grazing angle and the bound of the azimuthal angle. It decides the blind region that cannot be observed by the Moon-based SAR within the scope from the upper bound of the geocentric angle.

B. Examination of the Moon-Based SAR's Spatial Coverage

Now, the spatial coverage of the Moon-based SAR can be examined in terms of the coverage area and ground coverage. To begin with, the coverage area of the Moon-based SAR within the bound of (23) is detailed below.

1) Coverage Area on the Earth's Surface: Once the spatial coverage is limited to the range in (23), the Moon-based SAR's coverage area on the Earth's surface can be approximated by

$$\begin{aligned} S_c = 4R_E^2\{&\cos^{-1}[\Re(\phi_b)\cdot\kappa(\psi_{\text{far}}, R_{\text{SAR}})\cdot\zeta(\psi_{\text{far}}, R_{\text{SAR}})] \\ &- \cos^{-1}[\Re(\phi_b)\cdot\kappa(\psi_{\text{far}}, R_{\text{SAR}})\cdot\zeta(\psi_{\text{near}}, R_{\text{SAR}})] \\ &+ \sin^{-1}[\zeta(\psi_{\text{far}}, R_{\text{SAR}})] - \sin^{-1}[\zeta(\psi_{\text{near}}, R_{\text{SAR}})]\} \end{aligned} \tag{27}$$

with

$$\begin{cases} \Re(\phi_b) = \cos\phi_b - \sin\phi_b \\ \zeta(\psi_{\text{far}}, R_{\text{SAR}}) = [1 + \kappa^2(\psi_{\text{far}}, R_{\text{SAR}})]^{-0.5} \\ \zeta(\psi_{\text{near}}, R_{\text{SAR}}) = [1 + \kappa^2(\psi_{\text{near}}, R_{\text{SAR}})]^{-0.5}. \end{cases} \tag{28}$$

Accordingly, the ratio of the Moon-based SAR's coverage area to the global area can be written as

$$\begin{aligned} r_c = 4R_E^2\{&\cos^{-1}[\Re(\phi_b)\cdot\kappa(\psi_{\text{far}}, R_{\text{SAR}})\cdot\zeta(\psi_{\text{far}}, R_{\text{SAR}})] \\ &- \cos^{-1}[\Re(\phi_b)\cdot\kappa(\psi_{\text{far}}, R_{\text{SAR}})\cdot\zeta(\psi_{\text{near}}, R_{\text{SAR}})] \\ &+ \sin^{-1}[\zeta(\psi_{\text{far}}, R_{\text{SAR}})] - \sin^{-1}[\zeta(\psi_{\text{near}}, R_{\text{SAR}})]\} \\ &/\pi \times 100\%. \end{aligned} \tag{29}$$

The detailed derivations of (27) and (29) can be found in Appendix C.

According to (27) and (29), the coverage area on the Earth's surface is related to the ratio of the Earth's radius to the Earth–Moon-based SAR distance, the near- and far-grazing angles, and the bound of the azimuthal angle. As the bounds of the grazing and azimuthal angles are ascertained in the preceding analysis, the coverage area of the Moon-based SAR depends only on the ratio of the Earth's radius to the distance from the Earth to the Moon-based SAR. To further rigorously analyze the aspects of the coverage area, we plot the ratio of the Moon-based SAR's coverage area to the global area in Fig. 7 as a function of the Earth–Moon-based SAR distance.

From Fig. 7 we observe that the coverage area of the Moon-based SAR is positively correlated with the Earth–Moon-based SAR distance. Besides, the coverage area on the Earth's surface ranges from 21.01% to 21.08% of the global area. As the average eccentricity of the lunar orbit is 0.0549 [34], [35], thus, the variation in the Earth–Moon-based SAR distance is in the order of millions of meters during one period of the lunar revolution. In this respect, it can be concluded that the effect of the Earth–Moon-based SAR

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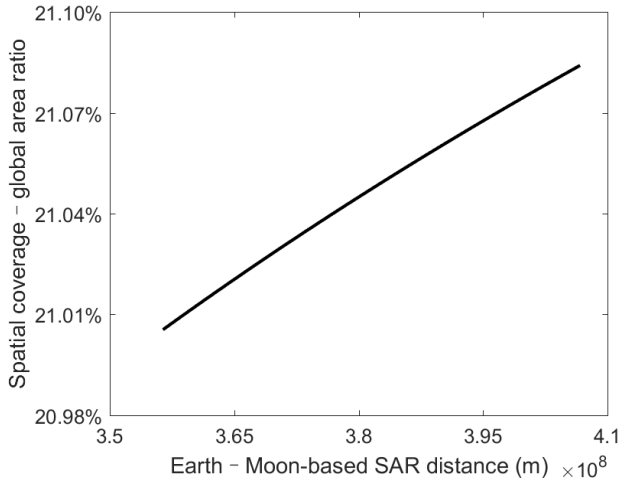


Fig. 7. Spatial coverage–global area ratio as a function of the Earth–Moon-based SAR distance.

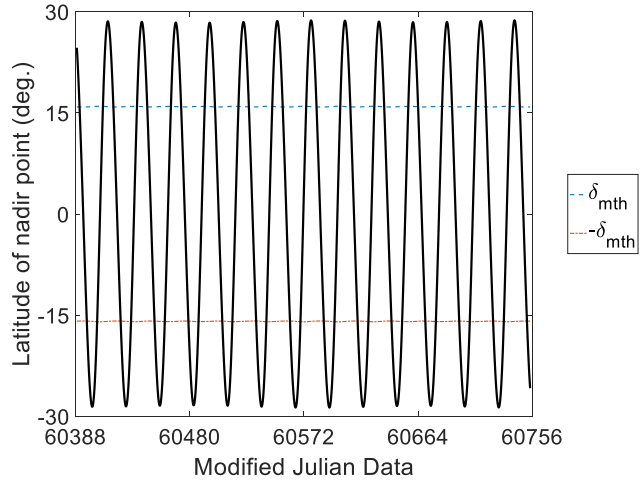


Fig. 8. Temporal varying latitude of the nadir point versus the threshold of the nadir point's latitude within epoch of 1 year.

distance is relatively small in determining the coverage area of the Moon-based SAR.

2) Latitudinal and Longitudinal Ground Coverages: In view of the imaging geometry of the Moon-based SAR, the maximum ground coverage's latitude occurs at 0° of the longitudinal deviation. Under this circumstance, the maximum and minimum latitudes of the ground coverage can be obtained from (23) and (24), as given by

$$\delta_g = \pm \cos^{-1}[\kappa(\psi_{\text{far}}, R_{\text{EM}})] + \delta_m. \tag{30}$$

By noting that the peak values of the Earth's latitude are ±90°, a threshold for the nadir point's latitude is defined

$$\delta_{m\text{th}} = 90^\circ - \cos^{-1}[\kappa(\psi_{\text{far}}, R_{\text{EM}})]. \tag{31}$$

When the grazing angle of the Moon-based SAR is bounded by the far-grazing angle, the coverages of the North and South poles of the Earth are still pertinent to the latitude of the nadir point, threshold value $\delta_{m\text{th}}$, and opposite value of the threshold, $-\delta_{m\text{th}}$. For a more thorough analysis, we now plot the temporally varying nadir point's latitude versus the thresholds of

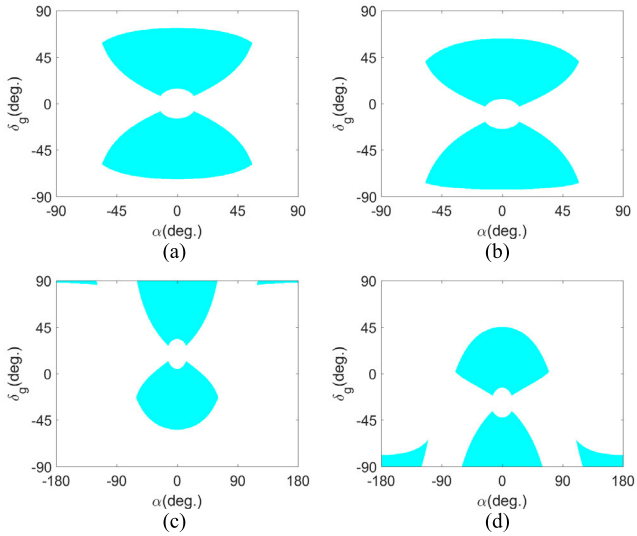


Fig. 9. Moon-based SAR's ground coverage with an Earth–Moon-based SAR distance of 385 000 km at the nadir point's latitude of (a) 0°, (b) −10°, (c) 18°, and (d) −28°.

$\delta_{m\text{th}}$ and $-\delta_{m\text{th}}$ within one Julian year (365.25 days) in Fig. 8. The initial visit time is set to 00:00:00 on March 20, 2024.

As shown in Fig. 8, once the nadir point's latitude is beyond the threshold of $\delta_{m\text{th}}$, the North Pole could be covered by the Moon-based SAR. Conversely, the South Pole would be covered by the Moon-based SAR when the nadir point's latitude is smaller than $-\delta_{m\text{th}}$. Under both conditions, the maximum longitudinal deviation can be up to as large as 180°. Regarding the Moon-based SAR with the nadir point's latitude standing at between $-\delta_{m\text{th}}$ and $\delta_{m\text{th}}$, neither the North Pole nor the South Pole of the Earth is observed by the Moon-based SAR. In this case, the maximum longitudinal deviation, being always smaller than 90°, takes the following form:

$$\alpha_{\max} = \sin^{-1}\{\cos\phi_b \cdot \cos^{-1}\delta_m \cdot [1 - \kappa^2(\psi_{\text{far}}, R_{\text{EM}})]^{-0.5}\}. \tag{32}$$

In Fig. 9, we plot the ground coverage of the Moon-based SAR at various latitudes of the nadir point (0°, −10°, 18°, and −28°) in the $\alpha - \delta_g$ domain. The Earth–Moon-based SAR distance is set to 385 000 km, the average distance from the Earth to the Moon-based SAR; it can be calculated that the threshold $\delta_{m\text{th}}$ for this distance is set to 15.916°.

As illustrated in Fig. 9, there are blind regions within the ground coverage bounded by the far-grazing angle due to the effects of the near-grazing angle and the bound of the azimuthal angle. Besides, the longitudinal ground coverage is rotationally symmetric about the axis of $\alpha = 0°$, whereas the latitudinal ground coverage is asymmetric except the nadir point being at the Earth's equator. Moreover, when the nadir point's latitude is smaller than the threshold of $\delta_{m\text{th}}$, neither of the Earth's Poles is covered by the Moon-based SAR. Meanwhile, the maximum size of the longitudinal deviation is always smaller than 90°. Regarding the Moon-based SAR's nadir point whose latitude is larger than $\delta_{m\text{th}}$, such as in Fig. 9(c), the North Pole is covered by the Moon-based SAR. However, the high latitudes in the southern hemisphere of the Earth cannot be observed. On the contrary, if the nadir

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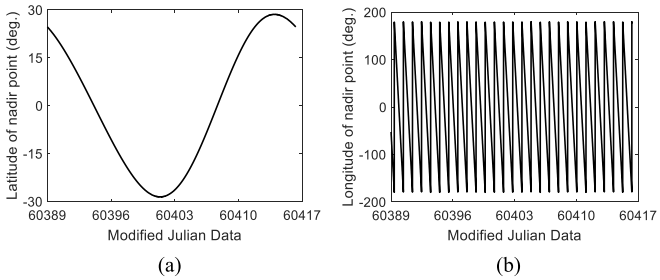


Fig. 10. During one period of the lunar revolution, temporal variations in the nadir point's (a) latitude and (b) longitude.

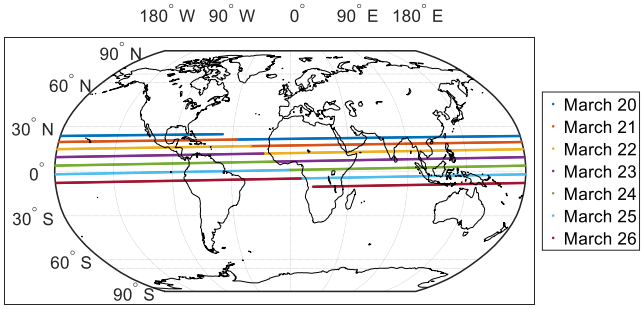


Fig. 11. Track of the Moon-based SAR's nadir point from March 20, 2024, to March 26, 2024. The synthetic aperture time is set to 200 s.

point's latitude is smaller than $-\delta_{\text{mth}}$, the South Pole can be covered by the Moon-based SAR, but the high latitudes in the northern hemisphere are not. Under both conditions, the maximum longitudinal deviation can reach as large as 180°.

C. Temporal Variations in the Spatial Coverage of the Moon-Based SAR

The spatial coverage of the Moon-based SAR is determined by the Earth–Moon-based SAR distance, the bounds of the grazing and azimuthal angles, and the position of the nadir point at a specified time. In addition, the nadir point, which temporally varies, effectively induces the temporal variation in the spatial coverage. The geographic coordinates of the nadir point, in light of the temporal variations, are discussed next.

The nadir point's latitude and longitude are provided by (6) and (7), both of which are a function of time and can be numerically obtained when the lunar ephemeris is applied. Now, the temporal variations in the latitude and longitude of the nadir point during one cycle of the lunar revolution are plotted in Fig. 10(a) and (b), respectively.

It can be observed from Fig. 10 that the latitude of the nadir point is mainly determined by the lunar revolution. By comparison, the nadir point's longitude is primarily decided by the Earth's self-rotation. Furthermore, the nadir point's longitude differs from that at the initial time after one period of the lunar revolution. This phenomenon is attributed to the Earth's self-rotation and lunar revolution around the Earth.

The temporal variation in the nadir point's geographical coordinate is a continuous process. However, the temporal variation in the spatial coverage is not exactly equivalent to that of the nadir point. A time interval related to synthetic

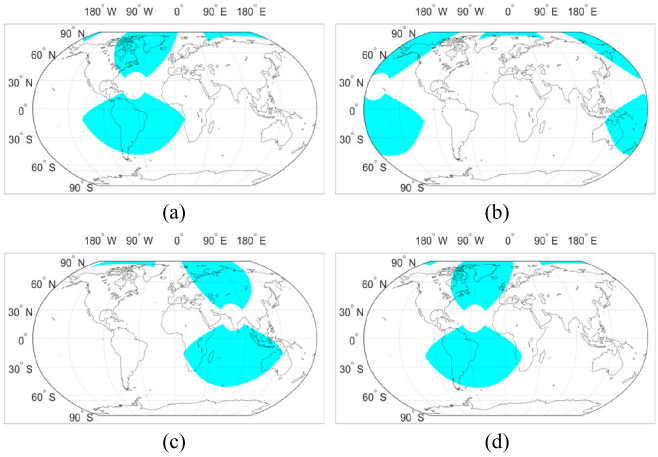


Fig. 12. Moon-based SAR's spatial coverage at (a) 00:00:00, (b) 08:00:00, (c) 16:00:00, and (d) 24:00:00 on March 20, 2024.

aperture, namely, the synthetic aperture time, still needs to be considered in the imaging formation of the coverage region. A synthetic aperture time of 200 s can realize an azimuth resolution around 10 m, which is suitable for the Earth observation from the Moon-based SAR [20]. Nevertheless, such a synthetic aperture time requires a large antenna with an aperture length in the azimuth larger than 1000 m, which poses a significant challenge to build in practice. The subaperture imaging technique provides a feasible method to implement the desired synthetic aperture time without using the large antenna [40], [41]. Thus, the synthetic aperture time of the Moon-based SAR may be set to 200 s.

Fig. 11 displays the track of the Moon-based SAR's nadir point within 1 week, from which we observe that the daily change in the nadir point's latitude is relatively small (around 3°). The temporal variation in the longitude of the nadir point within 1 day can exceed 340°. Hence, the temporally varying nadir point, together with the extensive spatial coverage, leads the revisit time to be less than 1 day for most of the regions covered by the Moon-based SAR.

IV. Numerical Simulations of the Spatiotemporal Coverage of the Moon-Based SAR

In this section, based on the lunar ephemeris from the DE430, numerical simulations are performed to complement the aforementioned analysis and to visualize the spatiotemporal coverage of the Moon-based SAR. We begin by discussing the hourly variations in the spatial coverage of the Moon-based SAR within 1 day.

A. Hourly Variations in the Spatial Coverage

Assume that the initial visit time is 00:00:00 on March 20, 2024, when the nadir point's latitude is relatively large. The hourly variations in the Moon-based SAR's spatial coverage (8-h time intervals) within 1 day are plotted in Fig. 12.

From Fig. 12, we see that most of the northern hemisphere of the Earth is measured by the Moon-based SAR, whereas the southern hemisphere of the Earth is less observed. Besides, it is understood that the spatial coverage moves from East to

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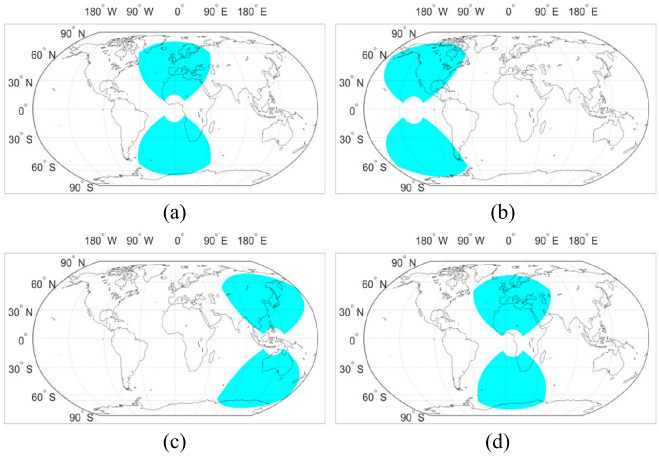


Fig. 13. Moon-based SAR's spatial coverage at (a) 00:00:00, (b) 08:00:00, (c) 16:00:00, and (d) 24:00:00 on March 25, 2024.

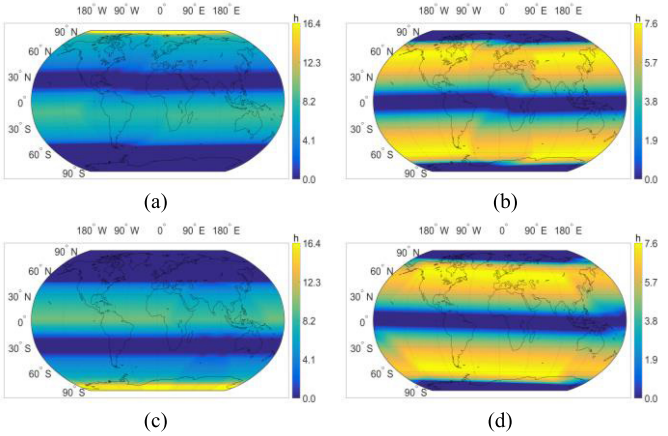


Fig. 14. Global daily accumulated visible times of the Moon-based SAR on (a) March 20, (b) March 25, (c) April 01, and (d) April 07, 2024.

West and the revisit time of the covered region is commonly less than 1 day, implying that the Earth's self-rotation is a dominant factor in determining the temporal variation in the spatial coverage. Comparing Fig. 12(d) with (a), we see that after a 1-day lapse, the spatial coverage has moved to the East. This phenomenon is caused by the lunar revolution around the Earth. Moreover, it can be observed that a large proportion of the Earth's surfaces are viewed (e.g., South America is almost entirely covered by the Moon-based SAR at 00:00:00). Nevertheless, there are still blind regions within the coverage bounded by the far-grazing angle. This suggests that the effects of the near-grazing angle and the bound of the azimuthal angle on the spatial coverage are significant. Hence, the spatial coverage is jointly determined by the near- and far-grazing angles and the bound of the azimuthal angle.

For comparison, the hourly variations in the spatial coverage of the Moon-based SAR with a small size of the nadir point's latitude are presented in Fig. 13. The visit time is reset to 00:00:00 on March 25, 2024, when the latitude of the nadir point approaches to 0°.

As shown in Fig. 13, the spatial coverage of the Moon-based SAR within 1 day is still significant when the size of the nadir point's latitude is small, but neither the North pole nor the South pole of the Earth can be observed. In addition, the revisit time of the observed region is less than 1 day as a result of the extensive spatial coverage and the Earth's self-rotation. Moreover, the effects of the lunar revolution result in the eastward motion of the spatial coverage after a 1-day lapse, as well. Furthermore, a blind region of considerable extent appears on the region bounded by the far-grazing angle, implying that the effects of the near-grazing angle and the bound of the azimuthal angle are indeed not negligible even when the size of the nadir point's latitude is small.

B. Global Accumulated Visible Time Within Different Periods

To exploit the spatiotemporal variation in the Moon-based SAR's ground coverage for a specified period, we analyze the accumulated visible time that is the accumulated time for a specific area covered by the Moon-based SAR within a specified period. The global daily accumulated visible time of the Moon-based SAR on March 20, March 25, April 01, and April 07 of 2024 are shown in Fig. 14(a)–(d), respectively.

As displayed in Fig. 14, the majority of the world can be covered by the Moon-based SAR within 1 day, and those regions are observed continuously, ranging from several hours to tens of hours. Besides, there is a conspicuous stripe that lies along the track of the nadir point, indicating that the daily accumulated visible time is affected by the near-grazing angle. In addition, the longitudinal accumulated visible time approximates to uniformly distribute. On the contrary, the distribution of the latitudinal accumulated visible time depends on the nadir point's latitude. When the nadir point's latitude is in the northern hemisphere of the Earth, the maximum latitudinal accumulated visible time appears at the region near the North Pole, whereas the fully Antarctica cannot be observed. The trend reverses when the nadir point lies in the southern hemisphere of the Earth. Once the nadir point's latitude approaches 0°, the latitudinal accumulated visible time trends to be symmetrically distributed. In such a situation, neither of the Earth's poles could be observed by the Moon-based SAR within 1 day.

One of the typical periods associated with the lunar revolution is the nodical month, which is the time interval between two successive transits of the Moon through the ascending node (or descending node). Accordingly, it is desirable to know the accumulated visible time of the Moon-based SAR within a nodical month. To this end, we plot the global accumulated visible time within an epoch from 00:00:00 on March 20, 2024, to 06:53:20 on April 16, 2024, in Fig. 15. The lunar orbital inclination with respect to the Earth's equator approximately peaks in this case.

Inspection of Fig. 15 indicates that the global nodical monthly accumulated visible time over a given epoch is in the order of several days. Besides, the latitudinal accumulated visible time is approximately symmetrically distributed about the Earth's equator, with its peak value occurring at the equator, $\pm 40°$ of latitudes and Earth's poles. Moreover, there are two stripes at $\pm 20°$ of latitudes, both of which correspond to the local minimum of the accumulated visible time. This phenomenon is given by the effect of the near-grazing angle. Furthermore, the longitudinal accumulated visible time

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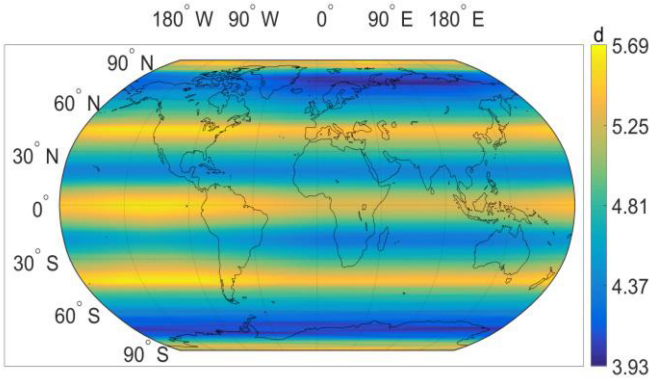


Fig. 15. Global accumulated visible time of the Moon-based SAR within an epoch from 00:00:00 on March 20, 2024, to 06:53:20 on April 16, 2024. The symbol "d" indicates days.

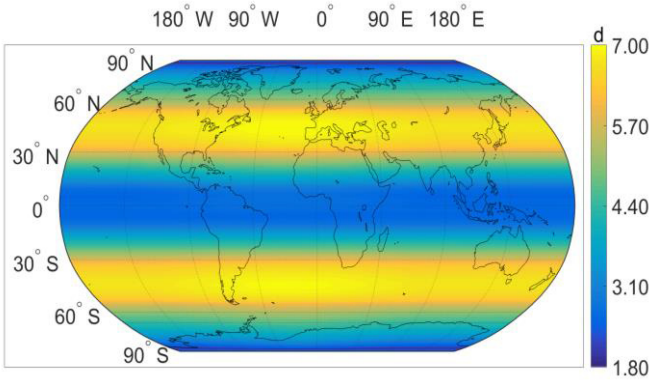


Fig. 16. Global accumulated visible time of the Moon-based SAR within an epoch from 00:00:00 on March 20, 2033, to 06:53:20 on April 16, 2033. The symbol "d" indicates days.

is uniformly distributed in the eastern hemisphere of the Earth. However, the distribution of the longitudinal accumulated visible time in the Earth's western hemisphere becomes inhomogeneous because there are peak regions around the longitude of 120° W.

To further analyze the spatiotemporal coverage of the Moon-based SAR within one nodical month when the orbital inclination is small, we simulate the accumulated visible time within an epoch from 00:00:00 on March 20, 2033, to 06:53:20 on April 16, 2033, in Fig. 16. The inclination angle between the lunar orbit and the Earth's equatorial plane in this epoch approaches to the minimum size.

It can be identified from Fig. 16 that the scope of the global nodical monthly accumulated visible time becomes wider for a small orbital inclination. In addition, the longitudinal accumulated visible time is approximately uniformly distributed. The distribution of the latitudinal accumulated visible time remains symmetrically distributed about the Earth's equator, but its peak value only appears at ±45° of latitudes. Furthermore, the accumulated visible time at low latitudes is dramatically reduced. Thus, the near-grazing angle shows a more significant impact on the spatiotemporal coverage when the inclination angle is small.

In general, there is not distinct regularity in the lunar revolution for one Julian year (365.25 days). Thus, the accumulated visible time over one Julian year cannot certainly reflect the tendency of the spatiotemporal coverage of the Moon-based SAR. As a result, the accumulated visible time of the Moon-based SAR within one Julian year is not presented. However, the inclination angle of the lunar orbit with respect to the Earth's equator periodically varies with a period of 18.6 years [32], [39]. Hence, the range of the latitude of the nadir point reaches a maximum or minimum every 18.6 years. Accordingly, it is of interest to illustrate the accumulated visible time of the Moon-based SAR over 18.6 years. By setting the initial visit time to 00:00:00 on March 20, 2024, the global accumulated visible time of the Moon-based SAR over 18.6 years is presented in Fig. 17.

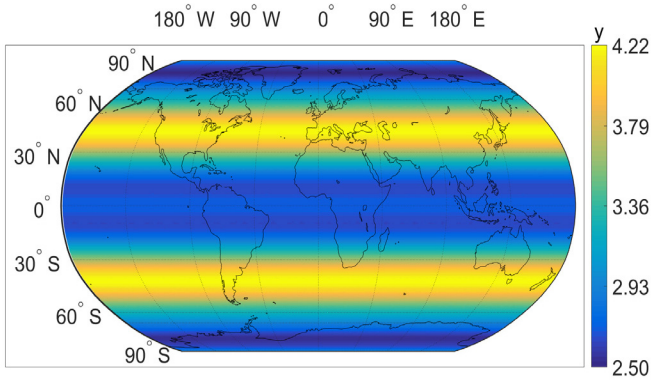


Fig. 17. Global accumulated visible time of the Moon-based SAR over a period of 18.6 years. The symbol "y" represents the Julian year.

According to Fig. 17, the global accumulated visible time of the Moon-based SAR over an 18.6-year period is in the order of several years. In addition, the longitudinal accumulated visible time trends to be uniformly distributed. In comparison, the latitudinal accumulated visible time is symmetrically distributed about the Earth's equator. What is more, the peak values of the latitudinal accumulated visible time appear at ±40° of latitudes, whereas the valleys occur at high latitudes with a size around 80°. Interestingly, there are apparent stripes above and below the Earth's equator in global accumulated visible time within an 18.6-year period, which is given by the effect of the near-grazing angle. This implies that the near-grazing angle exerts a considerable impact on the spatiotemporal coverage of the Moon-based SAR with a period of 18.6 years.

C. Optimal Site Selection of a Moon-Based SAR

To determine the optimal site of the Moon-based SAR for the Earth observation, we choose five candidates of the SAR system's site on the Moon's surface, for which the spatial coverage is not obscured by the lunar surface, as shown in Fig. 18. The coordinates of the five candidate sites for the Moon-based SAR are given in Table III. Note that the lunar libration is ignored, and the Moon is assumed to be a regular sphere in this analysis.

The rationales to choose these five sites are as follows. Sites M_2 and M_4 could maximize and minimize the latitudes of the nadir point of the Moon-based SAR, respectively. The nadir point's longitude peaks at the site of M_3, and it arrives at a valley value at the site of M_5. Finally, site M_1

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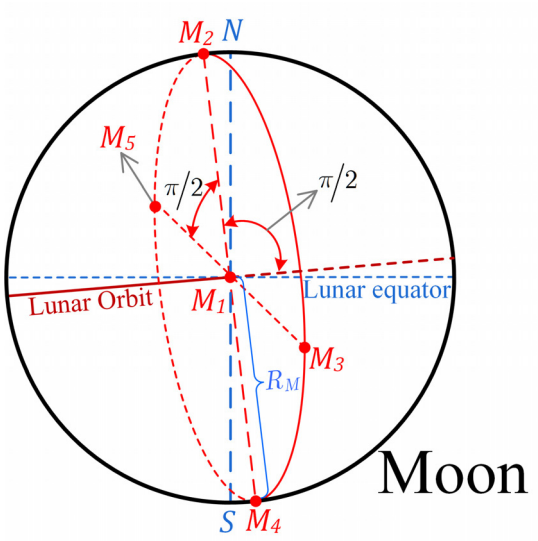


Fig. 18. Schematic of the Moon-based SAR's site, where M_1, M_2, M_3, M_4, and M_5 are the five candidates of the Moon-based SAR's site. M_1 is located at centers of the Moon. M_2, M_3, M_4, and M_5 are in the circle that is perpendicular to the lunar orbit and contains the selenocenter. R_M is the Moon's radius. N and S are the North and South poles of the Moon, respectively.

TABLE III
COORDINATES OF THE MOON-BASED SAR'S CANDIDATE SITES IN THE ECR

Coordinate / Site	Earth–Moon-based SAR distance	Longitude of the nadir point	Latitude of the nadir point
M_1	R_{SAR}	a_m	δ_m
M_2	$\sqrt{R_{SAR}^2 + R_M^2}$	a_m	$\delta_m + \tan^{-1}\left(\frac{R_M}{R_{SAR}}\right)$
M_3	$\sqrt{R_{SAR}^2 + R_M^2}$	$a_m - \tan^{-1}\left(\frac{R_M}{R_{SAR}}\right)$	δ_m
M_4	$\sqrt{R_{SAR}^2 + R_M^2}$	a_m	$\delta_m - \tan^{-1}\left(\frac{R_M}{R_{SAR}}\right)$
M_5	$\sqrt{R_{SAR}^2 + R_M^2}$	$a_m + \tan^{-1}\left(\frac{R_M}{R_{SAR}}\right)$	δ_m

R_M is the Moon's radius. R_{SAR}, a_m, and δ_m are, respectively, the distance from Earth to the Moon-based SAR, nadir point's longitude, and nadir point's latitude at the site M_i.

is chosen for comparison with the aforementioned analysis and previous analyses in [19] and [26]–[28]. Under such a configuration, the impacts of the Moon-based SAR's sites can be divided into the longitude and latitude aspects. Specifically, the influence of the longitude of the site can be investigated based on sites M_1, M_3, and M_5; by comparison, the effect of the site's latitude can be analyzed in accordance with sites M_1, M_2, and M_4.

Typically, the accumulated visible time over a period of one nodical month or one Julian year is different from that over the next period due to the temporal variation in the lunar orbital inclination. Thus, the accumulated visible time within one nodical month or one Julian year is not suitable for selecting the optimal site of a Moon-based SAR. In contrast, the lunar orbital inclination periodically varies every 18.6 years, and

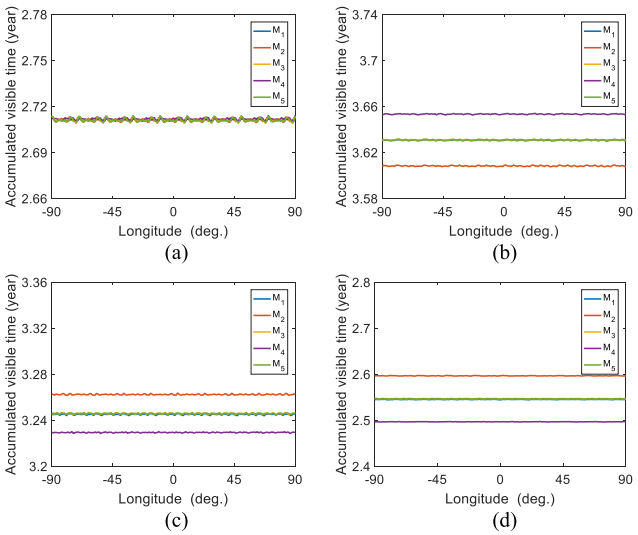


Fig. 19. 18.6-yearly longitudinal accumulated visible time at the Earth's latitude of (a) 0°, (b) 30° N, (c) 60° N, and (d) 80° N.

thus the accumulated visible time of the Moon-based SAR within a period of 18.6 years is approximately the same as that in the next period of 18.6 years. Consequently, to optimize the site selection, we investigate the accumulated visible time over 18.6 years using the five candidate sites depicted in Fig. 18. We plot the 18.6-yearly longitudinal accumulated visible time at various Earth's latitudes (0°, 30° N, 60° N, and 80° N) in Fig. 19. The initial time is set to 00:00:00 on March 20, 2024.

As presented in Fig. 19, the longitudinal accumulated visible time is approximately uniformly distributed wherever the Moon-based SAR is located. Besides, there is a tiny difference in the longitudinal accumulated visible time for the five Moon-based SAR's sites at the Earth's equator. As long as the other Earth's latitudes are concerned, the longitudinal accumulated visible time at site M_1 is still similar to that at sites M_3 and M_5. For example, at sites M_1, M_3, and M_5, the Moon-based SAR is able to view the Earth's equator with a mean of 71.32 days per year at the latitude of 30° N regardless of the Earth's longitude. Thus, the longitude of the nadir point exerts little impact on the accumulated visible time with a period of 18.6 years. In contrast, the longitudinal accumulated visible time at site M_1 is entirely different from that at sites M_2 and M_4. For instance, at site M_2, the Moon-based SAR can view the Earth's latitude of 80° N with a mean accumulated visible time of 49.01 days per year in spite of the Earth's longitude. By comparison, the mean accumulated visible time for the same latitude is around 50 days per year at site M_1 (or M_3, or M_5), while it increases to 51.04 days per year at site M_4. This indicates that the accumulated visible time is affected by the latitude of the nadir point.

Next, we explore how the nadir point's latitude impacts the accumulated visible time over an 18.6-year period in accordance with sites M_1, M_2, and M_4. By so doing, we plot the 18.6-yearly latitudinal accumulated visible time at various Earth's longitudes (0°, 60° E, 120° E, and 180° E) in Fig. 20 with an initial time of 00:00:00 on March 20, 2024.

Fig. 20 presents that the latitudinal accumulated visible time at different longitudes has the same tendency. In contrast, the

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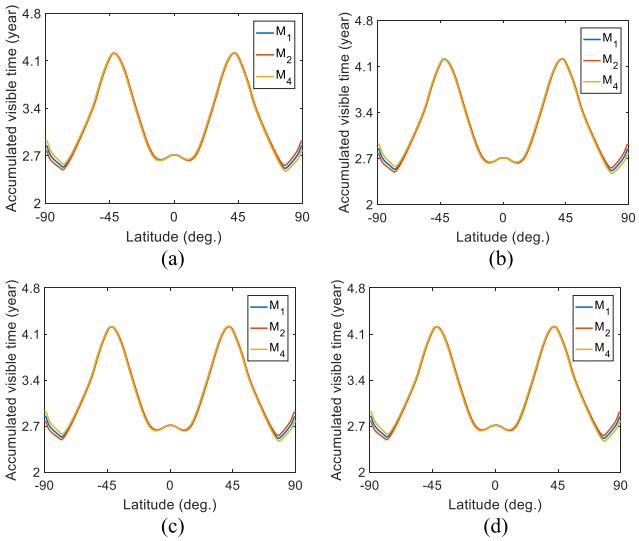


Fig. 20. 18.6-yearly latitudinal accumulated visible time at the Earth's longitude of (a) 0°, (b) 60° E, (c) 120° E, and (d) 180° E.

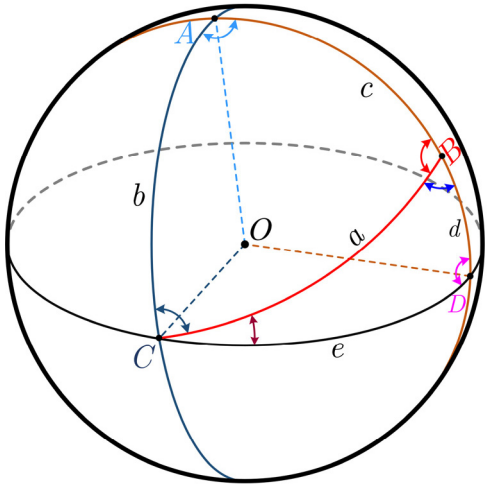


Fig. 21. Two adjacent auxiliary unit triangles, where a, b, and c represent the sides of the triangle ABC while a, d, and e stand for the sides of the triangle BCD. a is the public side of the two adjacent spherical triangles.

latitudinal accumulated visible time varies according to the Earth's latitude. For a Moon-based SAR locating at site M_1, it can be observed that the latitudinal accumulated visible time is the maximum with an average of 82.73 days per year at the Earth's latitude of 42° (N and S), while the latitudinal accumulated visible time is the minimum having an average of 49.31 days per year at latitudes of 78° N and 78° S. What is more, the location of the Moon-based SAR exerts a direct effect on the latitudinal accumulated visible time at high latitudes: When the Moon-based SAR is situated at M_2, the site near the North pole of the Moon, the latitudinal accumulated visible time in the Earth's northern hemisphere is longer than that in the Earth's southern hemisphere, especially at high latitudes. Conversely, the tendency reverses when the Moon-based SAR lies at M_4, the site near the South Pole of the Moon. If the Moon-based SAR is located at M_1, the region near the lunar equator, the latitudinal accumulated visible time is rotational symmetrical distributed with respect to the Earth's equator.

The Moon-based SAR can view the Arctic (66.57° N–90° N) for a specified longitude with a mean accumulated visible time ranging from 50.18 to 64.06 days per year at site M_2. In contrast, the mean latitudinal accumulated visible time for the Arctic at site M_4 is stands at between 48.52 and 63.41 days per year. Accordingly, it may be preferred to place the Moon-based SAR on the region near the site M_2 for observing the Arctic with longer available accumulated visible time. Conversely, the region around site M_4 on the Moon may be a good choice for measuring Antarctica (66.57° S–90° S) because the latitudinal accumulated visible time at site M_4 is always larger than that at site M_2 for a specified position at Antarctica. For the other regions, the site of the Moon-based SAR is of no consequence because the accumulated visible times over an 18.6-year period at different sites are very close to each other for most regions on the Earth.

V. Conclusion

In this article, the spatiotemporal coverage of a Moon-based SAR is investigated based on its imaging geometry. It is found that the spatial coverage of the Moon-based SAR is related to the bounds of the grazing and azimuthal angles, as well as the geographic coordinates of the nadir point at a specified time. Besides, the temporal variation in the spatial coverage is pertinent to the temporally varying nadir point of the Moon-based SAR. For the effective range-limited grazing and azimuthal angles, the far-grazing angle plays a dominant role in determining the spatiotemporal coverage of the Moon-based SAR. The effects of the near-grazing angle and the bound of the azimuthal angle are also profoundly significant to affect the spatiotemporal coverage. Regarding the optimal site selection, the site near the lunar North Pole is preferred for observing the Arctic of the Earth, while the site near the lunar South Pole may be a good choice for observing Antarctica. For other regions on the Earth, the site of the Moon-based SAR has no bearing on the spatiotemporal coverage of the Moon-based SAR over a long period. Consequently, the Moon-based SAR has the capacity to perform long-term, continuous Earth observations on a global scale to enhance our capabilities of understanding the planet.

Before closing, we note that it is of interest to conduct a comparative analysis on the coverage performance of the Moon-based SAR and the spaceborne SAR. We will pursue this subject in the future study.

Appendix A
Transformation From the ECI to the ECR

According to (4), the transformation from the ECI to the ECR can be achieved by applying the following matrix [31]:

$$\mathbf{U}_{\mathrm{ECR}}^{\mathrm{ECI}} = \Pi\Theta\mathbf{N}\mathbf{P}. \tag{A1}$$

The matrix $\mathbf{P}$ represents the coordinate change because of the precession and is written as follows:

$$\mathbf{P} = \mathbf{R}_z(-z)\mathbf{R}_y(\vartheta)\mathbf{R}_z(-\xi) \tag{A2}$$

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where ξ, ϑ, and z are all the auxiliary angles that describe precession in units of seconds. They are expressed as

$$\begin{cases} \xi = 2306''.2182T + 0''.30188T^2 + 0''.017998T^3 \\ \vartheta = 2004''.3109T + 0''.42665T^2 + 0''.014183T^3 \\ z = 2306''.2182T + 1''.09468T^2 + 0''.018203T^3. \end{cases} \quad \text{(A3)}$$

where T is the TDB time in Julian centuries from J2000.0. $\mathbf{R}_x$, $\mathbf{R}_y$, and $\mathbf{R}_z$ are the rotation matrices about the x-, y-, and z-axes, respectively. They can be expressed as

$$\mathbf{R}_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & \sin\theta \\ 0 & -\sin\theta & \cos\theta \end{bmatrix}$$
$$\mathbf{R}_y(\theta) = \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix}$$
$$\mathbf{R}_z(\theta) = \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}. \quad \text{(A4)}$$

The transformation matrix from the mean equator and equinox of the Earth to the true equator and equinox of the Earth can be written as

$$\mathbf{N} = \mathbf{R}_x(-\varepsilon - \Delta\varepsilon)\mathbf{R}_z(-\Delta\psi)\mathbf{R}_x(\varepsilon) \quad \text{(A5)}$$

where ε is the mean obliquity of the ecliptic, and it can be expressed as follows:

$$\varepsilon = 84381''.448 - 46''.815T - 0''.00059T^2 + 0''.001813T^3 \quad \text{(A6)}$$

$\Delta\varepsilon$ and $\Delta\psi$ in (A4) are the nutation in longitude and obliquity, respectively, and are, keeping the consistency with the DE430, given by

$$\begin{cases} \Delta\psi = -17''.206262\sin\Omega \\ \Delta\varepsilon = 9''.205348\cos\Omega. \end{cases} \quad \text{(A7)}$$

where Ω is the mean longitude of the ascending node of the lunar orbit measured on the ecliptic plane from the mean equinox of date, and it is

$$\Omega = 125°02'40''.280 - 1934°08'10''.549T + 7''.445T^2 + 0''.008T^3. \quad \text{(A8)}$$

The matrix Θ is related to the Earth's self-rotation, which takes the form of

$$\Theta = \mathbf{R}_z(\text{GAST}) \quad \text{(A9)}$$

with

$$\text{GAST} = \text{GMST} + \psi\cos\varepsilon \quad \text{(A9a)}$$

where GAST represents the Greenwich Apparent Sidereal Time, and GMST is the Greenwich Mean Sidereal Time, which can be obtained by

$$\text{GMST} = 24110^s.54841 + 8640184^s.812866 \cdot T_0 + 1.002737909350795 \cdot UT1 + 0^s.093104 \cdot T_u^2 - 0^s.0000062 \cdot T_u^3 \quad \text{(A10)}$$

where T_0 and T_u are given by

$$\begin{cases} T_0 = [JD(0^hUT1) - 2451545]/36525 \\ T_u = [JD(UT1) - 2451545]/36525. \end{cases} \quad \text{(A11)}$$

The time unit of T_0 and T_u is Universal Time1 (UT1), which is another time scale [31].

Finally, the matrix related to the polar motion of the Earth is given by

$$\Pi = \mathbf{R}_y(\Phi_y)\mathbf{R}_x(\Phi_x) \quad \text{(A12)}$$

where Φ_x and Φ_y are computed with an estimated linear correction in the DE430, as

$$\begin{cases} \Phi_x = \Phi_{x0} + 100T \cdot d\Phi_x/dt \\ \Phi_y = \Phi_{y0} + 100T \cdot d\Phi_y/dt. \end{cases} \quad \text{(A13)}$$

where Φ_{x0} and Φ_{y0} are the rotations of the x-axis and y-axis at J2000.0, respectively. $d\Phi_x/dt$ and $d\Phi_x/dt$ are the linear corrections rates of Φ_x and Φ_y, respectively. They are

$$\begin{cases} \Phi_{x0} = 0''.005675420332289347 \\ \Phi_{y0} = -0''.01702265691498953 \\ d\Phi_{x0}/dt = 0''.0002768991557448355\,yr^{-1} \\ d\Phi_{y0}/dt = -0''.001211859121655924\,yr^{-1}. \end{cases} \quad \text{(A14)}$$

APPENDIX B
DERIVATION OF THE PARAMETERS RELATED TO THE COVERAGE PERFORMANCE BASED ON THE IMAGING GEOMETRY

The position vectors of the Moon-based SAR and the ground target in the ECR are expressed by

$$\mathbf{R}_{\text{SAR}} = [R_{\text{SAR}}\cos\delta_m\cos a_m,\ R_{\text{SAR}}\cos\delta_m\sin a_m,\ R_{\text{SAR}}\sin\delta_m]^T \quad \text{(A15)}$$
$$\mathbf{R}_g = [R_E\cos\delta_g\cos a_g,\ R_E\cos\delta_g\sin a_g,\ R_E\sin\delta_g]^T. \quad \text{(A16)}$$

The slant range vector of the Moon-based SAR can be obtained by

$$\mathbf{R}_c = \mathbf{R}_{\text{SAR}} - \mathbf{R}_g. \quad \text{(A17)}$$

Thus, the slant range of the Moon-based SAR is

$$R_c = |\mathbf{R}_c| = \left[R_{\text{SAR}}^2 + R_E^2 - 2R_{\text{SAR}}R_E \times (\cos\delta_m\cos\delta_g\cos\alpha + \sin\delta_m\sin\delta_g)\right]^{0.5}. \quad \text{(A18)}$$

where $\alpha = a_g - a_m$. Hence, the elevation angle can be written as

$$\theta_{el} = \cos^{-1}\left\{\frac{\langle\mathbf{R}_c, \mathbf{R}_{\text{SAR}}\rangle}{|\mathbf{R}_c| \cdot |\mathbf{R}_{\text{SAR}}|}\right\} \quad \text{(A19)}$$

where "$\langle,\rangle$" represents the inner product. Substituting (A15) and (A17) into (A19), we obtain

$$\theta_{el} = \cos^{-1}\left\{\frac{R_{\text{SAR}} - R_E(\cos\delta_m\cos\delta_g\cos\alpha + \sin\delta_m\sin\delta_g)}{R_c}\right\}. \quad \text{(A20)}$$

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Similarly, the grazing angle is defined as

$$\psi = \sin^{-1}\left\{\frac{\langle \mathbf{R}_c, \mathbf{R}_g\rangle}{|\mathbf{R}_c|\cdot|\mathbf{R}_g|}\right\}. \tag{A21}$$

The grazing angle can be obtained by substituting (A15)–(A17) into (A21)

$$\psi = \sin^{-1}\{[R_{\mathrm{SAR}}(\cos\delta_m\cos\delta_g\cos\alpha + \sin\delta_m\sin\delta_g) - R_E]/R_c\}. \tag{A22}$$

The geocentric angle is expressed by

$$\theta_e = \cos^{-1}\left\{\frac{\langle \mathbf{R}_g, \mathbf{R}_{\mathrm{SAR}}\rangle}{|\mathbf{R}_g|\cdot|\mathbf{R}_{\mathrm{SAR}}|}\right\}. \tag{A23}$$

From (A15) and (A16), one can re-express (A23) as

$$\theta_e = \cos^{-1}(\cos\delta_m\cos\delta_g\cos\alpha + \sin\delta_m\sin\delta_g). \tag{A24}$$

The normal direction of the ground plane is

$$\mathbf{u}_{g_\mathrm{NOR}} = [\cos\delta_g\cos a_g, \cos\delta_g\sin a_g, \sin\delta_g]^T. \tag{A25}$$

The range vector in the ground range plane, upon projection, can be presented by

$$\mathbf{R}_{gc} = \mathbf{R}_c\cdot\left(\mathbf{I} - \mathbf{u}_{g_\mathrm{NOR}}^T\mathbf{u}_{g_\mathrm{NOR}}\right) \tag{A26}$$

where $\mathbf{I}$ is a unit matrix, with the setting of the Earth's orientation (Eastward) to the direction of 0° of the azimuthal angle, and the unit vector along the direction of the ground plane target's velocity vector is

$$\mathbf{u}_{vg} = \frac{\mathbf{K}\times\mathbf{R}_g}{|\mathbf{K}\times\mathbf{R}_g|} \tag{A27}$$

where "×" represents the cross product, and $\mathbf{K}$ indicates the vector of the Earth's rotation pointing to the North pole.

The azimuthal angle can be obtained by

$$\phi = \cos^{-1}\left\{\frac{\langle \mathbf{R}_{gc}, \mathbf{u}_{vg}\rangle}{|\mathbf{R}_{gc}|\cdot|\mathbf{u}_{vg}|}\right\}. \tag{A28}$$

By substituting (A15), (A16), and (A25)–(A27) into (A28), the azimuthal angle now takes the following form:

$$\phi = \cos^{-1}\{\cos\delta_m\sin\alpha[1 - (\cos\delta_g\cos\delta_m\cos\alpha + \sin\delta_g\sin\delta_m)^2]^{-0.5}\}. \tag{A29}$$

Appendix C
Area of the Moon-Based SAR's Spatial Coverage on the Earth's Surface

To examine the coverage area, we introduce two adjacent spherical triangles ΔABC and ΔBCD, both generated by the intersection of three great circles on a unit sphere, as demonstrated in Fig. 21.

The law of cosines states that [47]

$$\cos(c+d) = \cos b\cos e + \sin b\sin e\cos\angle ACD \tag{A30}$$

$$\cos\angle CBD = \sin\angle BCD\sin\angle ADC\cos e - \cos\angle BCD\cos\angle ADC. \tag{A31}$$

By the law of sines for spherical triangles, we have

$$\frac{\sin e}{\sin\angle CAD} = \frac{\sin b}{\sin\angle ADC} = \frac{\sin(c+d)}{\sin\angle ACD} \tag{A32}$$

where $\angle ACD = 90°$.

Now, assuming the latitude of the nadir point to be 0°, (A24) is reduced to

$$\theta_e = \cos^{-1}(\cos\delta_g\cos\alpha). \tag{A33}$$

As a result, (18) can be expressed as

$$\cos\delta_g\cos\alpha \approx \kappa(\psi, R_{\mathrm{SAR}}). \tag{A34}$$

Let the latitude with 0° of the longitudinal deviation in (A34) be b, and the longitudinal deviation with 0° of the latitude be e. It follows that

$$\begin{cases}\cos b \approx \kappa(\psi, R_{\mathrm{SAR}}),\\ \cos e \approx \kappa(\psi, R_{\mathrm{SAR}}).\end{cases} \tag{A35}$$

As a result, one can obtain

$$\begin{cases}\sin b = \sqrt{1-\kappa^2(\psi, R_{\mathrm{SAR}})}\\ \sin e = \sqrt{1-\kappa^2(\psi, R_{\mathrm{SAR}})}.\end{cases} \tag{A36}$$

According to the existing conclusions in (A30)

$$\cos(c+d) = \cos b\cos e + \sin b\sin e = \kappa^2(\psi, R_{\mathrm{SAR}}) \tag{A37}$$

Then, the sine of the side $(c+d)$ can be expressed as

$$\sin(c+d) = \sqrt{1-\kappa^4(\psi, R_{\mathrm{EM}})}. \tag{A38}$$

In accordance with the law of sines in (A32), one can obtain

$$\begin{cases}\sin\angle ADC = [1+\kappa^2(\psi, R_{\mathrm{EM}})]^{-0.5}\\ \sin\angle CAD = [1+\kappa^2(\psi, R_{\mathrm{EM}})]^{-0.5}.\end{cases} \tag{A39}$$

and the cosine of the angle $\angle ADC$ is

$$\cos\angle ADC = \kappa(\psi, R_{\mathrm{EM}})\cdot[1+\kappa^2(\psi, R_{\mathrm{EM}})]^{-0.5}. \tag{A40}$$

Substituting (A40) into (A31), we have

$$\cos\angle CBD = \kappa(\psi, R_{\mathrm{EM}})\cdot(\sin\angle BCD - \cos\angle BCD)\cdot[1+\kappa^2(\psi, R_{\mathrm{EM}})]^{-0.5} \tag{A41}$$

$$\cos\angle ABC = \cos(180° - \angle CBD) = \kappa(\psi, R_{\mathrm{EM}})\cdot(\cos\angle BCD - \sin\angle BCD)\cdot[1+\kappa^2(\psi, R_{\mathrm{EM}})]^{-0.5}. \tag{A42}$$

Now, the following relationships can be obtained:

$$\begin{cases}\angle ABC = \cos^{-1}\left\{\kappa(\psi, R_{\mathrm{EM}})\dfrac{\cos\angle BCD - \sin\angle BCD}{\sqrt{1+\kappa^2(\psi, R_{\mathrm{EM}})}}\right\}\\ \angle BAC = \angle CAD = \sin^{-1}\{[1+\kappa^2(\psi, R_{\mathrm{EM}})]^{-0.5}\}\\ \angle ACB = \angle ACD - \angle BCD = 90° - \angle BCD.\end{cases} \tag{A43}$$

Let $\angle BCD$ be ϕ_b, that is, the bound of the azimuthal angle, then the area of the spherical triangle ABC is [48]

$$\begin{aligned}S_{\mathrm{ABC}} &= \angle ABC + \angle BAC + \angle ACB - \pi\\ &= \sin^{-1}[(1+\kappa^2(\psi, R_{\mathrm{EM}}))^{-0.5}] - \phi_b - \pi/2\\ &\quad + \cos^{-1}[(\cos\phi_b - \sin\phi_b)\cdot\kappa(\psi, R_{\mathrm{EM}})\\ &\quad\cdot(1+\kappa^2(\psi, R_{\mathrm{EM}}))^{-0.5}].\end{aligned} \tag{A44}$$

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According to (14) and (23), the grazing and azimuthal angles are bounded by

$$\begin{cases} \psi_{\text{far}} \le \psi \le \psi_{\text{near}} \\ |\cos\phi| \le \cos\phi_b. \end{cases} \tag{A45}$$

By the symmetry of the spatial coverage with respect to the nadir point, the area of the Moon-based SAR's spatial coverage is

$$\begin{aligned} S_c = 4R_E^2\{&\cos^{-1}[(\cos\phi_b - \sin\phi_b)\cdot\kappa(\psi_{\text{far}}, R_{\text{EM}}) \\ &\cdot(1+\kappa^2(\psi_{\text{far}}, R_{\text{EM}}))^{-0.5}] \\ &- \cos^{-1}[(\cos\phi_b - \sin\phi_b)\cdot\kappa(\psi_{\text{near}}, R_{\text{EM}}) \\ &\cdot(1+\kappa^2(\psi_{\text{near}}, R_{\text{EM}}))^{-0.5}] \\ &+ \sin^{-1}[(1+\kappa^2(\psi_{\text{far}}, R_{\text{EM}}))^{-0.5}] \\ &- \sin^{-1}[(1+\kappa^2(\psi_{\text{near}}, R_{\text{EM}}))^{-0.5}]\}. \end{aligned} \tag{A46}$$

Finally, the ratio of the spatial coverage area to the global area can be approximately written as

$$\begin{aligned} r = \{&\cos^{-1}[(\cos\phi_b - \sin\phi_b)\cdot\kappa(\psi_{\text{far}}, R_{\text{EM}}) \\ &\cdot(1+\kappa^2(\psi_{\text{far}}, R_{\text{EM}}))^{-0.5}] \\ &- \cos^{-1}[(\cos\phi_b - \sin\phi_b)\cdot\kappa(\psi_{\text{near}}, R_{\text{EM}}) \\ &\cdot(1+\kappa^2(\psi_{\text{near}}, R_{\text{EM}}))^{-0.5}] \\ &+ \sin^{-1}[(1+\kappa^2(\psi_{\text{far}}, R_{\text{EM}}))^{-0.5}] \\ &- \sin^{-1}[(1+\kappa^2(\psi_{\text{near}}, R_{\text{EM}}))^{-0.5}]\}/\pi \times 100\%. \end{aligned} \tag{A47}$$

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